

1. Alex: A Study Companion for College Students with ADHD (IEEE Xplore)

Overview

- ADHD affects the academic, social and mental well-being of young adults. Socially assistive robots (SARs) have shown promise for people with ADHD, but most existing systems focus on children and lack productivity tools for college students.
- The authors built *Alex*, a study companion that helps with task prioritisation, scheduling and maintaining focus during work sessions.
- In a study with 15 university students who self-reported ADHD symptoms, Alex's support reduced feelings of overwhelm and improved focus. Twelve participants said they would use it again.

Project Objective

- To provide a productivity-oriented robot for college students (rather than children) that supports planning and sustained focus.

Research Gap

- Alex acts on what the user tells it; it does not observe what the user is actually doing at the desk, so it cannot detect distraction, phone use or absence.
- The paper is a short report (5 pages) with a small sample and short exposure, so long-term effects are unknown. (*Only the abstract was accessible to this review.*)

Research Question

- Can a social robot help college students with ADHD prioritise tasks, plan their time and keep focus during work sessions?

Key Contributions

- Early evidence that a robot companion can reduce overwhelm and improve focus in students with ADHD symptoms.
- A functional template for a student productivity robot: prioritisation, scheduling and focus sessions.
- Positive user acceptance (12 of 15 would reuse it).

Relevance to FocusBot

- Provides the functional baseline (focus sessions, scheduling, timers). FocusBot adds what Alex lacks: automatic, camera-based recognition of the user's activity.

Link: <https://ieeexplore.ieee.org/document/10974238>

2. Social Robot Coaching for Productivity Skill Development (IEEE)

Overview

- Productivity apps and self-help books are popular but fall short: books give generic, non-interactive guidance and apps do not teach organisational skills. Human coaching is personal but resource-intensive and hard to scale.
- The paper presents a proof-of-concept SAR that acts as an educational productivity coach. It delivers six lessons on time management and task prioritisation.
- Users interact through a chat interface and the robot answers in speech (which can be toggled). A dashboard monitors progress, mood, engagement, confidence per lesson and time spent per lesson, and offers personalised insights to foster reflection.
- Fifteen college students evaluated it, giving a System Usability Scale (SUS) score of 79.2 and high ratings for experience and engagement.

Project Objective

- To show that a SAR can deliver scalable, personalised productivity coaching and help students build time-management and prioritisation skills.

Research Gap

- The robot learns about the user mainly through the chat interface and self-report; it does not sense behaviour automatically.
- It is a proof-of-concept with no long-term study of whether skills or habits actually change.

Research Question

- Can a socially assistive robot serve as an effective, usable educational coach for productivity skills in college students?

Key Contributions

- A coaching architecture combining structured lessons, speech interaction and a progress dashboard.
- Evidence of good usability (SUS 79.2) and engagement with 15 students.
- A demonstration that analytics (mood, engagement, time per lesson) can support self-reflection.

Relevance to FocusBot

- Supports FocusBot's tracking dashboard. FocusBot can populate similar analytics automatically with detected study, break, distraction and idle time instead of relying on self-report.

Link : <https://doi.org/10.1109/RO-MAN63969.2025.11217634> · Preprint:
<https://arxiv.org/abs/2512.01105>

3. AMIGUS: A Robot Companion for Students (IEEE Xplore)

Overview

- Created during the pandemic, when students studied at home. Some students are isolated and need a push to study or interact with others.
- AMIGUS is a social robot offering companionship, motivational messages and tasks during online classes.
- Public descriptions of the project say the user reports mood with buttons (happy, sad, anguished, upset), expressive eyes on screens aim to build empathy, the head and back carry touch sensors, and the robot doubles as a stand for the user's phone. It was built by two mechatronics students at PUCP (Lima), advised by the university's psychological well-being office.

Project Objective

- To give lonely or isolated students a motivating companion that supports their mood and study routine.

Research Gap

- Mood is entered manually; the robot does not recognise what the student is doing, so its motivation is not tied to observed behaviour.
- It is the oldest paper in this review (2022), and *only the abstract and secondary descriptions were accessible*.

Research Question

- Can a low-cost expressive social robot motivate students and support their well-being while they study at home?

Key Contributions

- An early low-cost desk-robot design combining expressive eyes, touch sensing, motivational messages and emotional check-ins.
- Evidence that student-built, affordable hardware can support well-being goals.

Relevance to FocusBot

- A hardware and expressiveness precedent for FocusBot's display, lights and desk form factor. FocusBot replaces manual mood buttons with vision-based recognition.

Link: <https://dl.acm.org/doi/10.5555/3523760.3523852>

4. In-Dorm Socially Assistive Robot Schoolwork Companion for College Students with ADHD

Overview

- Students with ADHD adjust less easily to college and have lower GPAs and graduation rates. Earlier studies of simple SARs were limited to short in-lab sessions.
- The authors deployed a Blossom robot, a webcam and a Raspberry Pi with a 7-inch touch screen in students' dorm rooms. Sessions were 25-minute countdowns following the Pomodoro technique.
- The robot performed hard-coded idle motions (gaze shifts, posture sway, sighs) at random intervals that did not depend on the user's behaviour. The concept builds on "body doubling", where another presence in the room helps a person stay on task.
- A three-week within-subjects study with 11 students compared a no-robot week, a required-use week with the robot, and a voluntary-use week.
- Results: mean SUS 83.86; 91% (10 of 11) used the robot voluntarily when use was optional; no measurable change in executive-function scores or workload. Students reported companionship, accountability and better-structured time. 73% disliked the motor noise and 55% found the jerky movements distracting.
- Participants' suggestions: 73% wanted the robot to monitor attention and detect distraction; 82% wanted assistant features (reminders, calendar, assignment tracking). 73% said being recorded changed their behaviour (for example, not using their phones).

Project Objective

- To evaluate the usability of, and desire for, a long-term in-dorm SAR study companion, and to collect feedback on its design and features.

Research Gap

- The robot ignored the user's state (random motions only), and participants explicitly asked for distraction sensing and assistant functions.
- Small sample, possible novelty and experimenter-demand effects, and noisy actuators.

Research Question

- RQ1: Will students with ADHD find a SAR useful as a study companion? RQ2: What features do they like and dislike? RQ3: What features should be added?

Key Contributions

- Long-term in-dorm deployment evidence for study-companion robots.
- Framing of SARs as body doubles providing ambient companionship.
- A concrete, user-derived feature list (monitoring, reminders, quieter and more portable hardware).

Relevance to FocusBot

- The strongest motivation for FocusBot: users asked for exactly the camera-based awareness and assistant behaviour FocusBot provides. It also informs design choices: quiet, smooth actuation, a compact desk footprint and a 25/5-minute Pomodoro default.

Link: <https://arxiv.org/abs/2401.06289>

5. Motivating Students' Self-Study with Goal Reminder and Emotional Support

Overview

- Peer support helps students stay focused, but social anxiety and other barriers lead some to study alone. Self-study requires self-regulation, and students are pulled away by instant-reward distractions such as social media. Apps like Forest lack embodied presence and social interaction.
- A NAO robot acted as a peer study companion during a two-hour self-study session. It first guided each student to set a SMART goal, then "read" next to them on a tablet.
- Seventy-six students were recruited (74 analysed) across three conditions: *control* (presence only), *goal reminder* and *emotional support*. A human operator controlled all robot behaviour and triggered interventions when students looked distracted (dozing, using a phone, frustration, looking around).
- Results: both intervention conditions were rated easier to use than the control; goal reminders gave the highest willingness to reuse. Of 77 interventions delivered while students were distracted, 61 (79%) helped them refocus. Students who treated the robot as a "social other" were more likely to achieve their goals.

Project Objective

- To learn how task-oriented goal reminders and positive emotional support from a robot affect students' perceived focus, productivity and engagement during self-study.

Research Gap

- Interventions were delivered by a human wizard, not by a system that senses distraction itself; the setting was a single lab session where novelty may have inflated satisfaction.
- The authors conclude the robot must choose the right intervention for each student's momentary needs and traits, but no autonomous way of doing so was built.

Research Question

- How do goal-reminder and emotional-support behaviours from a robot study companion influence students' focus, productivity and engagement compared with presence alone?

Key Contributions

- Evidence that robot interventions during detected distraction help students refocus about 75–84% of the time.

- A contrast between goal reminders (accountability, a regulatory role) and emotional support (belonging, intrinsic motivation).
- Evidence that perceiving the robot as a social agent predicts goal achievement.

Relevance to FocusBot

- Directly justifies context-aware instead of fixed reminders. FocusBot can automate the wizard's role: camera-based distraction detection decides *when* to intervene and the Gemini API decides *what* to say, switching between reminder and encouragement styles.

Link: <https://arxiv.org/abs/2510.23860>

6. Understanding LLM-Powered Human-Robot Interaction

Overview

- LLMs promise natural, open-ended conversation for robots, but little was known about the design requirements of LLM-powered robots, which may differ from text and voice assistants and vary by task and context.
- A user study with 32 participants compared an LLM-powered Pepper robot against text-based and voice-based agents, all using GPT-3.5, across four conversational tasks: choose, generate, execute and negotiate.
- Findings: the robot raised expectations for sophisticated non-verbal cues and excelled in connection-building and deliberation (execute and negotiate tasks). It was less favoured in choose and generate tasks because of verbose answers, logical errors and social pressure from its gaze, which could induce anxiety. Voice-based interaction produced the most failures (interruptions, transcription errors).

Project Objective

- To identify design requirements for combining LLMs with robot embodiment, and the task contexts that benefit most from it.

Research Gap

- The robot had minimal non-verbal behaviour, there was no non-LLM robot baseline, quantitative results had high variance, and the study was short and laboratory-based.
- It did not examine proactive behaviour driven by sensing the user's activity.

Research Question

- How do people perceive LLM-powered robots, how do perceptions vary across task settings, and which task contexts benefit from robot embodiment?

Key Contributions

- Empirical comparison of text, voice and robot agents powered by the same LLM.
- Task-by-task evidence of where LLM robots help (rapport, deliberation) and where they hurt (precision, efficiency).
- Design implications: pair LLM speech with rich non-verbal cues, adapt the LLM to task characteristics, and guard against hallucination.

Relevance to FocusBot

- FocusBot's Gemini voice layer faces the same issues: expressive cues (lights, display, movement) should match the quality of its speech; replies should be short during focus work; and guardrails are needed against hallucinated or inappropriate responses.

Link: <https://arxiv.org/abs/2401.03217>

7. LLM-Powered Multi-Session Human-Robot Interactions with University Students

Overview

- An exploratory study of how open-domain, multi-session interaction with an LLM-powered social humanoid robot (EMAH) affects user perceptions and willingness to adopt it in a university setting.
- The conversation system supported multi-turn dialogue with person identification for long-term use. In the second session the robot opened by summarising the previous conversation, and participants could refer back to earlier topics.
- The authors conclude that LLM robots can sustain open-domain interaction over time, but that technical robustness, personalised conversation strategies and management of user expectations need improvement. *(Only the abstract, introduction and parts of the method were accessible to this review.)*

Project Objective

- To understand perceptions and adoption willingness for LLM-powered robots over repeated sessions with university students.

Research Gap

- Conversation was open-domain and not grounded in what the user was physically doing; the robot had no activity-aware, proactive behaviour.

Research Question

- How do multi-session interactions with an LLM-powered robot affect students' perceptions and willingness to adopt it?

Key Contributions

- Evidence on long-term LLM-robot interaction, including conversation memory across sessions.
- Practical lessons on robustness, personalisation and expectation management.

Relevance to FocusBot

- Supports giving FocusBot memory across sessions (past study and break patterns, preferred reminder style) and being transparent about its capabilities so expectations stay realistic.

Link: <https://doi.org/10.3389/frobt.2025.1585589>

8. FPI-Det: A Face – Phone Interaction Dataset for Phone-Use Detection

Overview

- Detecting phone use requires understanding behavioural context (how faces, hands and phones relate), not only recognising a phone. Generic benchmarks such as COCO do not capture these fine-grained human-device interactions.
- FPI-Det contains 22,879 images with 29,279 face and 10,255 phone annotations across workplace, education, transportation and public scenarios, with extreme scale variation (phones as small as 20×20 pixels), frequent occlusion and varied capture conditions.
- The authors benchmark YOLOv8, YOLOv11, DETR and Deformable DETR, and add a parameter-free rule that labels "phone in use" when a face and a phone are co-detected. YOLOv8-x gave the best phone detection (92.4% AP@50), YOLOv8-n was fastest (about 417 FPS on a V100 GPU), and classification accuracy was about 87–89%. Common failures: false positives on tablet-like objects and misses on small or occluded phones.

Project Objective

- To provide the first large-scale benchmark with joint face and phone annotations and to evaluate phone-use detection and classification.

Research Gap

- Analysis is per image, with no temporal reasoning, and the method struggles to separate merely holding a phone from actively using it.

Research Question

- How well can modern one-stage and transformer detectors localise faces and phones and infer phone use under diverse real-world conditions?

Key Contributions

- A new dataset and evaluation protocol for phone-use detection.
- Comparative baselines for YOLO and DETR families, showing that lightweight YOLO variants are fast enough for edge deployment.
- A simple co-detection rule for phone-use classification.

Relevance to FocusBot

- A ready baseline for FocusBot's phone-distraction module: a small YOLO model plus the co-detection rule. FocusBot should add temporal smoothing (flag phone use only after it persists for several seconds) to reduce false alarms.

Link: <https://arxiv.org/abs/2509.09111>

9. AI-Based Multimodal Biometrics for Detecting Smartphone Distractions

Overview

- Learners lose attention to internal factors (motivation), system-related factors (course design) and contextual factors such as smartphone use. Learning platforms often lack detailed behavioural data.
- The paper uses physiological signals and head pose to detect phone use during computer-based online learning. Single signals such as brain waves or heart rate gave limited accuracy; head pose alone reached 87%, and a multimodal model combining all signals reached 91%. *(Only the abstract was accessible; the full text is paywalled.)*

Project Objective

- To detect smartphone distraction during tasks that require sustained attention, with a focus on online learning.

Research Gap

- Biosensors are impractical for everyday desk use, and the paper discusses implications and limitations of real-time support rather than delivering a working intervention system.

Research Question

- Can multimodal biometric signals and head pose reliably detect phone use during online learning?

Key Contributions

- Evidence that head pose is a strong single cue (87%) and that fusing signals adds a further gain (91%).
- A multimodal learning-analytics approach to detecting phone-related distraction.

Relevance to FocusBot

- Justifies a camera-only design built on head pose and gaze, which captures most of the accuracy without wearable sensors.

Link : https://doi.org/10.1007/978-3-032-03870-8_3

10. Ethically-Aware Participatory Design of a Productivity Social Robot (IEEE)

Overview

- College students face academic and life stressors that affect productivity. Conventional productivity tools demand sustained self-discipline, which many students struggle with, leading to disruptive app-switching.
- Earlier productivity SARs did not involve students in their design. This study used participatory design: interviews with college students, and a collaborative workshop with students and a Student Success and Well-Being Coach, with ethical questions built into the process.

Project Objective

- To gather student-driven insights on productivity challenges and desirable features of a productivity-focused SAR, with ethics integrated into the design phase.

Research Gap

- The output is design insight and recommendations; no perception-driven working system was built or tested.

Research Question

- What productivity challenges do college students face, and what features and characteristics should a productivity robot have?

Key Contributions

- Comprehensive insights into student productivity challenges.
- SAR design preferences and actionable recommendations for robot characteristics.
- A model for embedding ethics in participatory design.

Relevance to FocusBot

- Camera monitoring raises privacy concerns, so FocusBot should adopt privacy-by-design: on-device processing, no raw video storage, a visible camera indicator and user-controlled monitoring level.

Link: <https://arxiv.org/abs/2512.01111>